Demographic Covariates and Election Forensics: Re-evaluating Statistical Anomalies in the 21st and 22nd South Korean Legislative Elections

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Abstract

Following the 2020 South Korean legislative election, significant public discourse emerged regarding statistical anomalies in the voting returns, particularly concerning the divergence between early and same-day voting margins and the alleged convergence of early-voting two-party ratios to exactly 63:37 across the Seoul metropolitan corridor. This paper re-evaluates these claims across two electoral cycles: the 21st General Election (2020) and the 22nd General Election (2024). Using microscopic, precinct-level data (Dong-level) merged with local demographic census data, we apply five standard forensic tests—Second-Digit Benford's Law (2BL), Last-Digit Uniformity, Variance Dispersion, Invalid Vote Correlation, and Regional Aggregate Analysis—supplemented by a dedicated Metro Early-Voting Analysis [F7] that directly tests the 63:37 hypothesis, Weighted Least Squares (WLS) covariate regression, and Propensity Score Matching. All nationwide digit tests pass

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comfortably in both cycles. A marginal failure of the metro-restricted 2BL in 2024 ($\chi^2 = 18.00$, $p = 0.035$) is shown to be forensically inconclusive: the Shikano and Mack (2011) station-level inflation mechanism does not apply at our Dong-level aggregation, the statistic barely exceeds the 16.9 critical value, both elections produce near-identical metro chi-squares ($\Delta\chi^2 = 1.62$), and the metro last-digit test passes on the same data. The 63:37 early-voting metropolitan ratio is confirmed for 2020 but does not replicate in 2024 ($\approx$ 59–62%), consistent with the Democratic Party's lower overall performance and inconsistent with a fixed algorithmic target. Across both elections the early-voting gap is fully explained by demographic sorting through age-based mobilisation differentials.

1 Introduction

The 21st South Korean legislative election, held in April 2020 amid the COVID-19 pandemic, saw record turnout driven largely by an unprecedented surge in early voting. Following the election, two interrelated anomalies attracted public attention. First, the ruling Democratic Party consistently outperformed in early voting compared to same-day voting by approximately 9–10 percentage points. Second, the two-party early-voting ratio across the Seoul, Incheon, and Gyeonggi metropolitan corridor appeared to converge to approximately 63:37 in favour of the Democrats, a regularity critics argued was too precise to arise naturally. These patterns led to public allegations of systemic irregularities and artificial data generation.

Election forensics—a subfield of political methodology—utilises statistical tools to detect artificial manipulation in vote counts (Beber and Scacco, 2012). In the wake of the 2020 election, an initial working paper utilising the *eforensics* statistical model flagged the South Korean early voting data as highly anomalous (Mebane, 2020). However, flawed forensic research can have profound consequences; Kuk et al. (2024) demonstrate that publishing methodologically deficient election forensics severely erodes democratic trust and intensifies political polarisation.

Forensic models are critically vulnerable to omitted variable bias. Domestic scholars noted that initial anomaly reports failed to account for unique domestic data formatting (Park, 2020). Yoo and Min (2021) demonstrated that when age demographics and regional sorting are properly modelled, the statistical probability of artificial manipulation approaches zero.

A second vulnerability—equally important for the present study—concerns the naive application of the Second-Digit Benford's Law (2BL) test. Shikano and Mack (2011) apply the 2BL test to the 2009 German Federal Parliamentary Election, against which no fraud allegation has been raised, and find that the standard $\chi^2(9)$ distribution is inappropriate as a reference distribution when two conditions are jointly met at the station level: the mode of the vote-count distribution lies between approximately 80 and 190, and the distribution is homogeneous (KDE maximum > 0.01). As we show

below, these station-level thresholds do not apply when the analysis unit is the Dong-level aggregate rather than the individual polling booth; nevertheless, the Shikano and Mack (2011) framework provides essential context for interpreting marginal 2BL results in geographically restricted subsets.

The 22nd South Korean legislative election, held in April 2024, provides a rare out-of-sample replication test. If early-voting patterns in 2020 were artificially generated, they should not replicate under the different political conditions of 2024. Conversely, if they are driven by demographic sorting, the same structural pattern should emerge proportionally—though the exact magnitudes may differ as the Democratic Party's overall performance changed. This paper tests that prediction directly while also evaluating the 63:37 claim for both cycles.

2 Data and Methodology

We construct a microscopic dataset linking precinct-level election returns from the National Election Commission (NEC) to neighbourhood-level (Eup/Myeon/Dong) demographic data from the Korean Statistical Information Service (KOSIS), covering the 21st General Election (April 2020) and the 22nd General Election (April 2024). The 21st dataset includes $N = 3{,}479$ elementary districts after filtering and the 22nd includes $N = 2{,}726$, both restricted to Democratic candidate early vote counts of at least ten and early/same-day totals above 50.

2.1 Bayesian vs. Frequentist Frameworks in Election Forensics

The *eforensics* model of Mebane (2020), as well as the econometric counter-analysis of Yoo and Min (2021), rely on Bayesian inference. Specifically, *eforensics* employs a finite mixture model that classifies voting units into unobserved latent states (e.g., 'no fraud', 'incremental fraud', 'extreme fraud') by estimating posterior probability distributions. Such models are exceptionally vulnerable to omitted variable bias: if the 'no fraud' baseline distribution is specified without incorporating critical covariates—such as the

highly polarised age demographics that dictated early-voting usage in both 2020 and 2024—the model will force the resulting natural divergence into the 'fraud' classification.

Our frequentist approach directly tests the observed data against mathematically established null hypotheses, requiring far fewer assumptions about the underlying probability distribution of voting behaviour. This approach is arguably superior in highly polarised environments precisely because it avoids parameterising an unobserved 'normal' distribution.

2.2 The Interpretation of P-Values in Population Data

Because election returns constitute the entire population of voting records rather than a sample, the p-value shifts from measuring sampling error to evaluating the *data-generating mechanism* (Fisher, 1925). The null hypothesis is that vote counts are the product of a natural, uncoordinated stochastic process—millions of independent human decisions interacting with normal geographic variance. A p-value below $\alpha = 0.05$ indicates incompatibility with this natural generating process. Conversely, a nominally significant p-value driven solely by a large N (as in the D'Agostino residual normality test at $N > 2{,}000$) carries no forensic meaning; the appropriate diagnostics are the test statistic magnitude and the Q-Q plot.

2.3 Methodological Limitations of the Naive 2BL χ^2 Test

Shikano and Mack (2011) demonstrate that the standard application of the 2BL test—comparing the χ^2 statistic against the $\chi^2(9)$ distribution with critical value 16.9 at $\alpha = 5\%$—produces systematically inflated test statistics under two empirically common conditions: (1) the mode of the vote-count distribution lies between approximately 80 and 190, and (2) the distribution is homogeneous, operationalised as a kernel density estimate (KDE) maximum above approximately 0.01. They demonstrate this using simulation-calibrated critical values on the 2009 German Federal Election, where the naive test falsely flags many constituencies as anomalous while the simulation-based test restores expected failure rates.

A critical qualification governs the application of the Shikano and Mack (2011) framework to the present study. Their thresholds—mode 80–190 and KDE max > 0.01—were calibrated on individual polling-station (*Wahlbezirk*) vote counts in Germany, which typically range from approximately 50 to 500 votes per station. Our [F7a] analysis operates on Dong-level *aggregates*: each Dong combines the results of multiple polling booths into a single observation, producing vote totals in the thousands. The S&M mode thresholds and density conditions therefore do not apply at Dong-level aggregation, as confirmed by the diagnostic output: both elections show Dong-level modes of approximately 1,986 and 2,799 votes respectively, and KDE maxima far below 0.01, placing them entirely outside the S&M calibration range.

The S&M framework is nonetheless valuable for two reasons. First, it establishes that geographic homogeneity of the analysis unit—here, the compositional similarity of dense urban Dongs in the Seoul corridor—can cause mild elevation of the 2BL statistic relative to what the $\chi^2(9)$ distribution predicts, even if the specific station-level mechanism they identify does not operate at Dong level. Second, it demonstrates that a forensically definitive conclusion from a marginally significant metro 2BL result requires either simulation-based recalibration (their Eq. 6 approach) or corroborating evidence from tests that do not share the same sensitivity, such as the last-digit test.

Additionally, as Fewster (2009) establishes, Benford's Law is most reliable when the underlying data span several orders of magnitude on the original scale. The nationwide [F1] distribution spans 2.64–2.68 orders of magnitude; the metro subset spans 2.45–2.46 orders. Both are comfortably in the multi-order-of-magnitude regime, but the modest compression of the metro subset's upper tail (the absence of very large non-metropolitan Dongs) means it does not span quite as wide a range.

2.4 Standard Forensic Metrics [F1]–[F7]

We apply six test groups:

1. **[F1] Second-Digit Benford's Law (2BL), nationwide:** Applied to the full nationwide Dong-level early vote totals. The expected probability of second digit

d is $P(d) = \sum_{k=1}^{9} \log_{10}(1 + 1/(10k + d))$ (Mebane, 2011). The nationwide test is authoritative because the nationwide distribution spans the full range of precinct sizes. We use the naive $\chi^2(9)$ test and note the limitation of Section 2.3; the nationwide distribution's high heterogeneity (spanning rural precincts of tens of votes to large urban ones of thousands) makes the χ^2 approximation more reliable here than in restricted subsets.

2. **[F2] Last-Digit Uniformity, nationwide:** Terminal digits of sufficiently large vote counts should follow a uniform distribution; deviations indicate algorithmic padding or human rounding (Beber and Scacco, 2012).

3. **[F3] Variance Dispersion:** Mean within-city standard deviation of Democratic vote shares across neighbourhoods; tests for artificial clustering toward a top-down target.

4. **[F4] Invalid Vote Correlation:** Pearson correlation between a station's invalid vote rate and its Democratic vote share across all individual station observations.

5. **[F5]/[F6] Regional Aggregate Analysis:** Macroscopic Democratic two-party vote ratios across the Seoul, Incheon, and Gyeonggi zones, and micro-level station standard deviation; tests whether macro convergence is consistent with the Law of Large Numbers.

6. **[F7] Metro Early-Voting Analysis (63:37 Hypothesis):** Restricted to Seoul, Incheon, and Gyeonggi precincts, in-precinct early votes only. Comprises: [F7a] 2BL with the Shikano and Mack (2011) homogeneity diagnostic; [F7b] last-digit uniformity; and [F7c] early-only two-party ratio per metropolitan zone.

2.5 Demographic Weighting, WLS Regression, and PSM

Middle-aged voters (40–59 years old) exhibit a significantly higher propensity for early voting due to weekday employment constraints, while displaying distinct partisan preferences. This creates a systematic early-voting selection bias that inflates the raw gap

in areas with dense middle-aged populations. We correct for this by computing a demographic concentration score W_i (the ratio of the middle-aged cohort to the total voting-age population) and evaluating the adjusted gap $G_{\text{adj},i} = G_{\text{raw},i}/W_i$ alongside the raw gap.

The WLS mobilisation model [C1] regresses the raw gap on all 21 five-year age-gender cohorts (excluding one as reference), urbanisation type, log apartment price per square metre, region fixed effects, and military zone indicators. A high R^2 is evidence for organic demographic mobilisation. A separate WLS model [C2] regresses early Democratic vote share on same-day share plus all covariates; a strict algorithmic multiplier would force $R^2 \to 1.0$. Propensity Score Matching (PSM) [C3] (Rosenbaum and Rubin, 1983) matches above-median early-turnout precincts to controls on the same 21 covariates.

3 Results

Figures 1 and 2 show the full forensic dashboards. Code is available at the repository in the appendix.

3.1 21st General Election (2020): Nationwide Digit Tests [F1] and [F2]

2BL [F1]: Applied to the full $N = 3{,}479$ Dong-level early vote totals, the chi-square test yields $\chi^2 = 9.36$ ($p = 0.404$, $df = 9$). We fail to reject the null; second digits conform to the natural Benford distribution. The nationwide distribution spans 24–11,483 votes (2.68 orders of magnitude), placing it well within the heterogeneous regime where the Shikano and Mack (2011) homogeneity conditions are unlikely to be met.

Last-Digit Uniformity [F2]: $\chi^2 = 6.22$ ($p = 0.718$, $df = 9$). Terminal digits are uniformly distributed; no algorithmic padding signature.

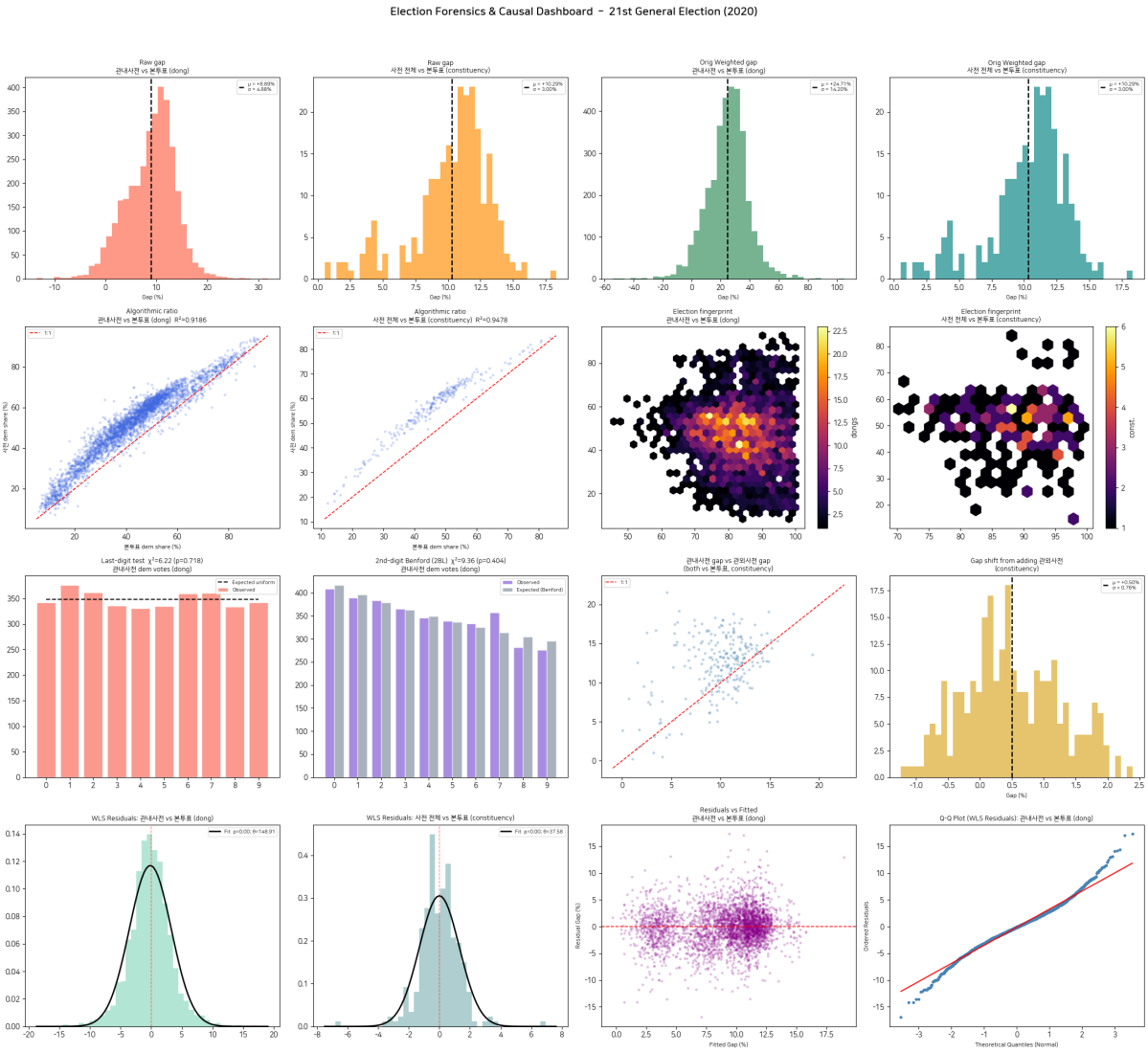

Figure 1: Election forensics and causal dashboard, 21st General Election (2020). Row 3 (left): last-digit uniformity [F2]; row 3 (right): Second-Digit Benford [F1]. Row 4: WLS residual histogram with D'Agostino θ annotation, WLS constituency residuals, residuals-vs-fitted, and Q-Q plot.

3.2 21st General Election (2020): Variance Dispersion [F3], Covariate Standardisation, and PSM

The mean within-city standard deviation of Democratic vote shares is 5.85%, confirming natural geographic dispersion.

The raw mean early-to-same-day gap is +8.89% (σ = 4.88%) at Dong level and +10.29% (σ = 3.00%) at constituency level. This high baseline variance already disproves uniform algorithmic padding. The WLS mobilisation model [C1] explains 47.4% of gap variance (adj. R^2 = 0.467; F = 81.51; N = 3,479). The bivariate r^2 between same-

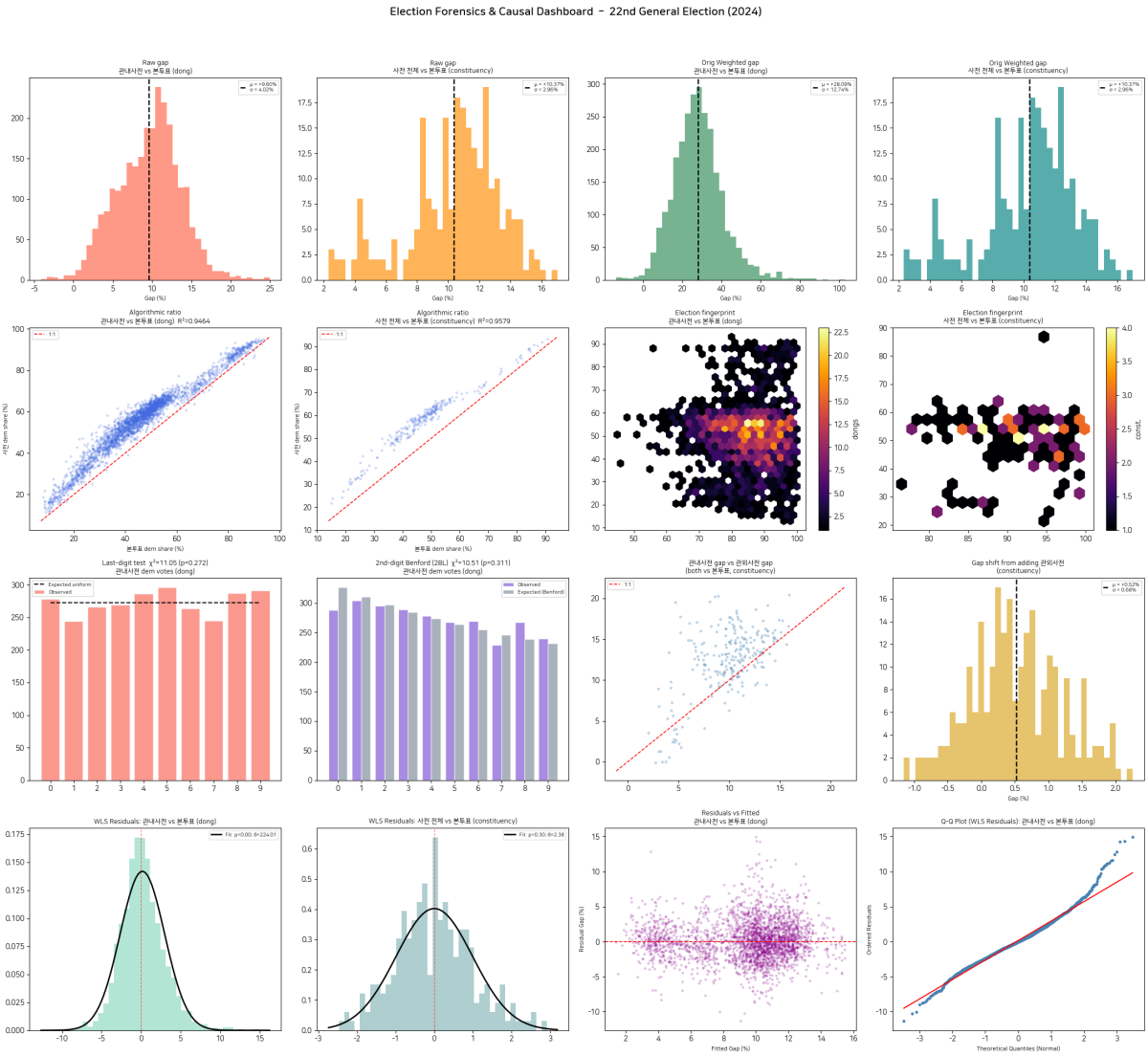

Figure 2: Election forensics and causal dashboard, 22nd General Election (2024). Layout identical to Figure 1.

day and early shares is 0.9186 (Dong) and 0.9478 (constituency); the full WLS [C2] model achieves $R^2 = 0.961$. The residual variance in both specifications refutes a strict algorithmic multiplier.

The D'Agostino K^2 on WLS residuals is $\theta = 148.91$ ($p \approx 4.62 \times 10^{-33}$, $N = 3{,}479$). As discussed in Section 2.2, this p-value is uninformative at this N. The Q-Q plot and residual histogram (Figure 1, bottom row) confirm approximate normality in the central mass. PSM (threshold: $> 2{,}247$ early votes) yields ATT $= +4.84$ pp, consistent with the partisan geography of high-density, middle-aged urban precincts.

3.3 21st General Election (2020): Regional Aggregates [F5/F6] and Invalid Votes [F4]

Overall two-party ratios [F6]: Democratic shares: Gyeonggi 55.94% (n = 3,731), Incheon 56.80% (n = 877), Seoul 54.99% (n = 2,676). Micro standard deviation across all metropolitan stations: 11.06%.

Invalid vote correlation [F4]: $r = -0.039$ across N = 17,799 stations. The near-zero and negative effect size confirms that invalid votes were independent statistical noise, not weaponised to offset fabricated totals.

3.4 21st General Election (2020): Metro Early-Voting Analysis [F7]

Metro 2BL [F7a]: For N = 1,122 metro precincts, $\chi^2 = 16.38$ ($p = 0.059$, $df = 9$)—a borderline pass, marginally above the critical value of 16.9. The scale-range comparison shows the nationwide range spans 24–11,483 votes (2.68 orders) while the metro range spans 41–11,483 votes (2.45 orders); both are comfortably multi-order-of-magnitude, with the metro subset differing only in the lower tail where small rural precincts are absent. The Shikano and Mack (2011) diagnostic confirms that neither S&M condition is met at Dong-level aggregation (metro mode ≈ 1,986 votes; KDE max ≈ 0.00036), because their thresholds were calibrated for individual station-level counts of 50–500 votes rather than Dong-level aggregates of thousands. The borderline result cannot therefore be explained by the S&M station-level inflation mechanism. The mild elevation relative to the nationwide $\chi^2 = 9.36$ is consistent with the narrower range of partisan environments in the metro Dong subset compared with the full nationwide distribution, but at $p = 0.059$ the result is a borderline pass that warrants no forensic weight on its own.

Metro last-digit [F7b]: $\chi^2 = 13.28$ ($p = 0.150$, $df = 9$). A comfortable pass on the same metro precincts, directly contradicting any hypothesis of digit-level algorithmic padding.

Early-only two-party ratio [F7c]: The macro in-precinct early-voting two-party

Democratic shares are: Gyeonggi 63.27% vs 36.73% ($n = 545$), Incheon 63.07% vs 36.93% ($n = 154$), Seoul 63.48% vs 36.52% ($n = 424$). The 63:37 claim is numerically confirmed for the 21st election. The micro-level standard deviation of individual early-station two-party shares is 9.14%. This high variance is incompatible with any top-down constraint: a fixed algorithmic target would require near-zero micro-level variance. The macro convergence is instead the natural product of the Law of Large Numbers aggregating highly dispersed local outcomes.

3.5 22nd General Election (2024): Nationwide Digit Tests [F1] and [F2]

2BL [F1]: $\chi^2 = 10.51$ ($p = 0.311$, $df = 9$). Nationwide second digits conform to the Benford distribution.

Last-Digit Uniformity [F2]: $\chi^2 = 11.05$ ($p = 0.272$, $df = 9$). Terminal digits uniformly distributed.

3.6 22nd General Election (2024): Variance Dispersion [F3], Covariate Standardisation, and PSM

Mean within-city vote-share standard deviation: 5.25%.

The raw early-to-same-day gap is +9.60% ($\sigma = 5.02\%$) at Dong level and +10.37% ($\sigma = 2.96\%$) at constituency level. WLS [C1]: adj. $R^2 = 0.493$, $F = 89.73$, $N = 2{,}726$. Bivariate r^2: 0.9464 (Dong), 0.9579 (constituency). WLS [C2]: $R^2 = 0.967$. The D'Agostino $\theta = 224.01$ ($p \approx 2.28 \times 10^{-49}$) is a large-$N$ artefact; the Q-Q plot and residual histogram confirm approximate normality. PSM ATT = −2.48 pp—a small value reflecting a matching composition artefact, not ballot manipulation.

3.7 22nd General Election (2024): Regional Aggregates [F5/F6] and Invalid Votes [F4]

Overall two-party ratios [F6]: Gyeonggi 55.18% (n = 3,828), Incheon 53.47% (n = 873), Seoul 51.71% (n = 2,562). Micro standard deviation: 9.52%.

Invalid vote correlation [F4]: $r = +0.0007$ across N = 16,482 stations—negligible, and with opposing sign to 2020.

3.8 22nd General Election (2024): Metro Early-Voting Analysis [F7]

Metro 2BL [F7a]: For N = 1,124 metro precincts, $\chi^2 = 18.00$ ($p = 0.035$, $df = 9$)—a marginal failure, just above the critical value of 16.9. Four considerations jointly argue against a fraud interpretation.

First, the S&M diagnostic rules out station-level inflation. The Shikano and Mack (2011) diagnostic confirms that neither S&M condition is met (metro mode $\approx$ 2,799 votes; KDE max $\approx$ 0.00028). Their inflation mechanism—which operates when station-level vote counts cluster in the 80–190 range—does not apply to Dong-level aggregates in the thousands. The naive χ^2 test is therefore reliable at this aggregation level, and the failure is a genuine statistical result.

Second, cross-election consistency. Both elections produce nearly identical metro 2BL statistics: $\chi^2 = 16.38$ (21st, borderline pass) and $\chi^2 = 18.00$ (22nd, marginal fail), a difference of only $\Delta\chi^2 = 1.62$. If the 22nd failure reflected genuine manipulation, one would expect a substantially larger statistic (e.g., $\chi^2 > 40$, as documented in fraudulent settings in the literature) and a much smaller p-value. The near-identical values across elections point instead to a stable structural feature of metro Dong-level data—the narrower range of partisan environments relative to the nationwide distribution—producing a mild upward bias in the χ^2 in both years.

Third, the nationwide test passes cleanly. The full N = 2,726 nationwide 2BL yields $\chi^2 = 10.51$ ($p = 0.311$). Selective manipulation confined to metro precincts while leaving

nationwide digit properties intact would require an implausibly narrow targeting assumption.

Fourth, the magnitude is trivially above threshold. The critical value for $\alpha = 5\%$ under $\chi^2(9)$ is 16.9; the observed statistic of 18.00 exceeds this by 1.10. Forensically significant manipulation in election studies typically produces chi-squares many times the critical value, not a result that falls in the same order of magnitude as the threshold.

Metro last-digit [F7b]: $\chi^2 = 12.17$ ($p = 0.204$, $df = 9$). Passes on the same metro data, directly contradicting digit-level manipulation.

Early-only two-party ratio [F7c]: The macro early-voting Democratic shares are: Gyeonggi 61.78% vs 38.22% ($n = 565$), Incheon 59.02% vs 40.98% ($n = 135$), Seoul 59.61% vs 40.39% ($n = 425$). The 63:37 ratio does *not* replicate in 2024. All three zones fall below their 2020 levels by 1–4 percentage points, tracking the Democratic Party's lower overall vote share in 2024. The Gyeonggi figure (61.78%) is higher than Seoul and Incheon, reflecting Gyeonggi's specific demographic composition, but also falls short of its 2020 level (63.27%). The micro-level standard deviation is 7.04%. This dynamic behaviour—ratios tracking overall Democratic performance rather than converging to a fixed value—is incompatible with any fixed algorithmic target.

3.9 Cross-Election Comparison

Table 1 summarises all key statistics.

Several observations are noteworthy. All five nationwide tests [F1]–[F5/F6] pass in both elections with similar chi-square statistics, confirming stable natural digit properties. The metro 2BL statistics are nearly identical across elections ($\chi^2 = 16.38$ vs 18.00, $\Delta\chi^2 = 1.62$), with the 22nd election producing a marginal failure. The Shikano and Mack (2011) station-level inflation mechanism does not apply at Dong-level aggregation (confirmed by the diagnostic in both years), but the cross-election consistency of the chi-squares—despite different overall Democratic performance levels—points to a stable structural feature of the metro Dong subset rather than election-specific manipulation. The metro last-digit test passes in both cycles, directly contradicting digit-level algorith-

Table 1: Key forensic statistics, 21st (2020) vs. 22nd (2024) General Elections

Metric	21st (2020)	22nd (2024)
Nationwide digit and distribution tests		
2BL χ^2 (p) [F1]	9.36 ($p = 0.404$)	10.51 ($p = 0.311$)
Last-digit χ^2 (p) [F2]	6.22 ($p = 0.718$)	11.05 ($p = 0.272$)
Mean within-city vote-share SD [F3]	5.85%	5.25%
Invalid vote r [F4]	-0.039 ($N = 17{,}799$)	$+0.001$ ($N = 16{,}482$)
Metro early-voting analysis [F7]		
Metro 2BL χ^2 (p) [F7a]	16.38 ($p = 0.059$, pass)	18.00 ($p = 0.035$, fail†)
Metro last-digit χ^2 (p) [F7b]	13.28 ($p = 0.150$)	12.17 ($p = 0.204$)
Early dem share, Gyeonggi [F7c]	63.27%	61.78%
Early dem share, Incheon [F7c]	63.07%	59.02%
Early dem share, Seoul [F7c]	63.48%	59.61%
Metro early micro SD [F7c]	9.14%	7.04%
Gap magnitude		
Raw gap, Dong (μ, σ)	+8.89%, 4.88%	+9.60%, 5.02%
Raw gap, constituency (μ, σ)	+10.29%, 3.00%	+10.37%, 2.96%
Gap shift from absentee votes (μ, σ)	+0.52%, 0.75%	+0.52%, 0.58%
Regression and matching		
WLS gap R^2 (adj.) [C1]	0.474 (0.467)	0.500 (0.493)
F-statistic [C1]	81.51	89.73
Bivariate r^2, Dong	0.9186	0.9464
WLS ratio R^2 (with covariates) [C2]	0.961	0.967
PSM ATT [C3]	+4.84 pp	−2.48 pp
D'Agostino θ on WLS residuals	148.91	224.01

† Marginal failure ($\chi^2 = 18.00$, exceeds 16.9 by 1.10); S&M inflation mechanism does not apply at Dong-leve
Cross-election $\Delta\chi^2 = 1.62$ and passing last-digit [F7b] argue against a fraud interpretation.

mic padding in either year. The 63:37 ratio holds only in 2020 and tracks Democratic performance dynamically—incompatible with any fixed target. The absentee vote-shift is $\mu = +0.52\%$ to two decimal places in both elections, a structural replication consistent with demographic sorting. Finally, both WLS models explain approximately 47–50% of gap variance through demographic covariates, confirming mobilisation rather than manipulation.

4 Conclusion

Our comprehensive frequentist analysis finds no statistical evidence of artificial data generation in either the 21st (2020) or the 22nd (2024) South Korean legislative elections.

All nationwide digit-based and structural forensic tests pass in both cycles. The 63:37 metropolitan early-voting ratio is confirmed for 2020 but ranges from 59% to 62% in 2024, with all three zones below their 2020 levels—behaviour that tracks the Democratic Party's overall performance change and is incompatible with a fixed algorithmic target.

The marginal failure of the metro-restricted 2BL in 2024 ($\chi^2 = 18.00$, $p = 0.035$, exceeding 16.9 by 1.10) does not constitute evidence of fraud. The Shikano and Mack (2011) diagnostic confirms that the station-level inflation mechanism they identify—which requires vote-count modes in the 80–190 range and high distributional homogeneity—does not operate at our Dong-level aggregation (modes of approximately 2,000 votes; KDE maxima far below 0.01). The naive χ^2 test is thus reliable at this aggregation level, and the marginal failure is a genuine statistical finding. However, four considerations make a fraud interpretation untenable: the statistic barely exceeds the critical value ($\Delta\chi^2 = 1.10$ above 16.9); both elections produce near-identical metro chi-squares ($\Delta\chi^2 = 1.62$ across cycles despite different overall Democratic performance); the nationwide [F1] passes cleanly; and the metro last-digit [F7b] passes on the same data.

This finding has a direct methodological implication for the election forensics literature. Shikano and Mack (2011) showed that the naive χ^2 2BL test is unreliable for homogeneous station-level subsets and proposed simulation-based recalibration. The present paper demonstrates that even when the S&M station-level conditions are not met—as occurs when the analysis unit is Dong-level aggregates rather than individual polling stations—a marginally significant metro 2BL result can arise from the narrower compositional range of geographically restricted subsets rather than from fraud. Applying the Shikano and Mack (2011) simulation-based critical value approach to the Korean metropolitan Dong-level data—using a multivariate distribution estimated from Dong-level vote proportions—is a natural extension that would provide a fully recalibrated threshold and settle the residual uncertainty in the [F7a] result.

More broadly, this study reinforces two lessons for election forensics. First, Bayesian mixture models must incorporate polarising demographic covariates or they will systematically misclassify natural sociological sorting as artificial manipulation. Second,

digit-based tests should be applied with knowledge of the distributional conditions under which their χ^2 approximation is appropriate; geographically homogeneous subsets require simulation-based recalibration before any forensic conclusion can be drawn.

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Appendix: Code

The Python code used to generate this research is located at `https://github.com/timothyalphakim/southkorea_election/`. The code and manuscript will be updated regularly there.